

Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{24.63 - 25}{\sigma} = -1.0364$	M1	This mark is given for standardising as part of a method to find σ
	$\sigma = 0.357$	A1	This mark is given for a correct value of σ
	$P(D > K) = 0.4$ or $P(D < K) = 0.6$	B1	This mark is given for
	$\frac{k - 25}{\sigma} = \frac{k - 25}{0.357} = 0.2533$	M1	This mark is given for using a normal model to find the probability
	$k = 25.09$	A1	This mark is given for a correct value for k
(b)	$Y \sim B(200, 0.45)$ so $W \sim N(90, 49.5)$	B1	This mark is given for setting up the normal distribution approximation of the binomial
	$P(Y < 100) \approx P(W < 99.5) = P\left(Z < \frac{99.5 - 90}{\sqrt{49.5}}\right)$	M1	This mark is given for using the normal model with a continuity correction
	$= 0.912$	A1	This mark is given for finding a correct value of the probability
(c)	$H_0 : \mu = 25$ $H_1 : \mu < 25$	B1	This mark is given for both hypotheses in terms of μ found correctly
	$\bar{D} \sim N\left(25, \frac{0.16^2}{20}\right)$	M1	This mark is given for a method to set up the normal distribution
	$P(\bar{D} < 24.94) = 0.0468$	A1	This mark is given for using the model to find a correct p -value
	$p = 0.0468 < 0.05$, so reject H_0	M1	This mark is given for a correct comparison and non-contextual conclusion
	There is sufficient evidence to support Hannah's belief	A1	This mark is given for a correct conclusion in context stated
(Total 13 marks)			

Q2.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$H_0 : \rho = 0$ $H_1 : \rho > 0$	B1	This mark is given for both hypotheses in terms of ρ found correctly
	For sample size 24 at the 5% level of significance, the critical value = 0.3438	M1	This mark is given for selecting a suitable critical value compatible with H_1
	$0.446 > 0.3438$, so reject H_0 There is evidence that the product moment correlation coefficient (pmcc) is greater than 0	A1	This mark is given for a correct conclusion stated
(b)	The value of the pmcc is close to 1 so there is a strong positive correlation	B1	This mark is given for a correct explanation about the strength of the correlation
(c)	$\log_{10} y = -1.82 + 0.89 \log_{10} x$	M1	This mark is given for a correct substitution of both c and m
	$y = 10^{-1.82 + 0.89 \log x}$	M1	This mark is given for dealing with logs to find an expression in terms of y
	$y = 10^{-1.82} \times 10^{0.89 \log x}$ $y = 10^{-1.82} \times 10^{(\log x)^{0.89}}$	M1	This mark is given for a method to find values for a and n
	$y = 0.015 \times x^{0.89}$	A1	This mark is given for find a correct value of $a = 0.015$
		A1	This mark is given for find a correct value of $n = 0.89$
(Total 9 marks)			